Hall Effect

{Eric Cao and Adriano Rasmussen} Physics 3600 - Advanced Physics Lab June 24, 2025

Northeastern University

Abstract

We explored the transport properties of doped silicon using four-wire resistance and Hall effect techniques. Resistivity values of 7.82 ± 0.21 and $7.54 \pm 0.20\ \Omega \cdot \mathrm{cm}$ were obtained from voltage drops across two regions. A thermoelectric test confirmed p-type behavior through hole diffusion. In a magnetic field, the sample exhibited a Hall voltage consistent with positive charge carriers. From the Hall slope, we calculated a carrier density of $(2.544 \pm 0.071) \times 10^{15}\ \mathrm{cm}^{-3}$ and mobilities of 313.74 ± 12.01 and $325.62 \pm 12.47\ \mathrm{cm}^2/\mathrm{V} \cdot \mathrm{s}$. These results agree with literature values for moderately doped p-type silicon.

Conclusion

This experiment used Hall effect and four-wire measurements to study electrical transport in doped silicon. The sample showed **p-type** behavior, based on the direction of the Hall voltage and results from the hot probe test.

The I–V curve showed asymmetry and nonlinear resistance, which could be related to **Schottky effects** or differences in contact quality, though we did not test this systematically.

The Hall voltage had a small offset, which might come from wiring differences or small thermal imbalances. Still, the slope gave reasonable values for carrier density and mobility in moderately doped silicon.

References

1. S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices*, 3rd ed. (Wiley, 2007).
2. R. F. Pierret, *Semiconductor Device Fundamentals* (Addison-Wesley, 1996).
3. E. H. Rhoderick and R. H. Williams, *Metal–Semiconductor Contacts*, 2nd ed. (Clarendon Press, 1982).
4. HyperPhysics, "Hall Effect" and "Seebeck Effect," http://hyperphysics.phy-astr.gsu.edu.
5. NIST, "Hall Effect Measurements," https://www.nist.gov/calibrations/hall-effect-measurements.

Motivation/Introduction

The Hall effect offers a powerful way to probe the internal charge transport of semiconductors. By applying a magnetic field perpendicular to the current, we can measure the Hall voltage and determine the type, density, and mobility of carriers. Real samples deviate from ideal behavior due to Schottky barriers and thermoelectric effects, which introduce nonlinearities and asymmetries in transport. This experiment combines Hall and four-wire resistance methods to investigate these effects in doped silicon.

Methods

Sample Prep: Silicon wafer with 6 indium contacts mounted to a pre-wired PCB. Fine wires were soldered using indium/lead-tin solder. All contact resistances were verified to be $< 1\ \mathrm{M}\Omega$.
Resistivity (4-wire): Current applied across contacts 1–4; voltage measured across 2–3 and 5–6. Resistivity was computed using:

$$\rho = R \cdot \frac{A}{L}$$

Thermal Probe: A soldering iron near contacts 1 or 4 induced a Seebeck voltage. The sign of the response confirmed **p-type** behavior via hole diffusion.
Hall Effect: A constant current I_S flowed through the sample while V_H was measured across 3–5 under a varying magnetic field B. Carrier density and mobility were extracted from:

$$n = \frac{I_S}{e\,d\left(\frac{dV_H}{dB}\right)}, \quad \mu = \frac{1}{ne\rho}$$

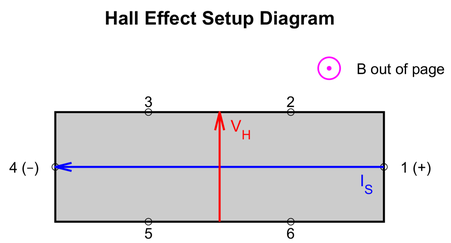

Figure 1: Hall setup: current (I), magnetic field (B), and Hall voltage (V_H) directions.

Electrical Characterization Results

I–V Behavior Forward conduction is near-linear; reverse conduction is nonlinear due to Schottky barrier suppression and enhanced carrier injection.

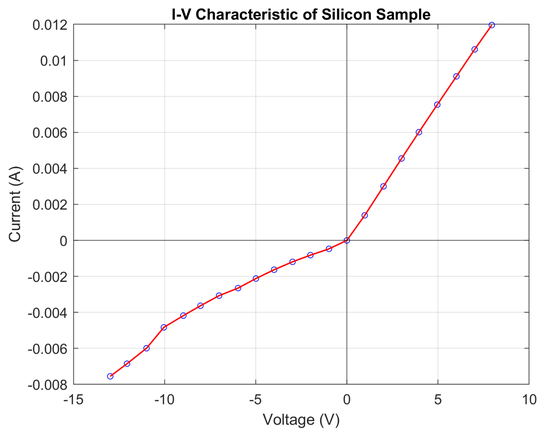

Figure 2: Current–voltage (I–V) characteristic. Reverse bias shows barrier-limited conduction.

Resistance vs. Voltage Reverse bias resistance decreases with increasing voltage, consistent with field-enhanced carrier transport.

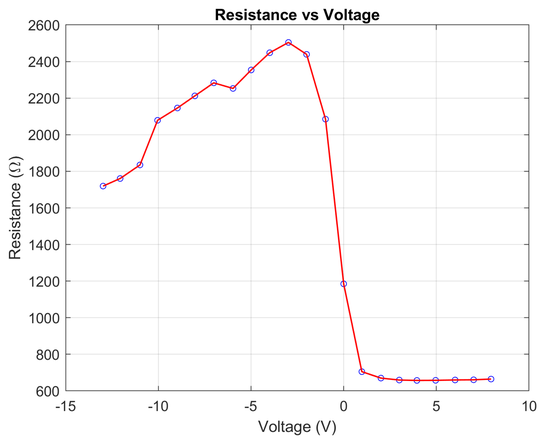

Figure 3: Resistance vs. voltage. Forward bias is flat; reverse shows field dependence.

Hall Voltage vs. Magnet Current A linear trend was observed; the slope was used to compute the Hall voltage vs. magnetic field.

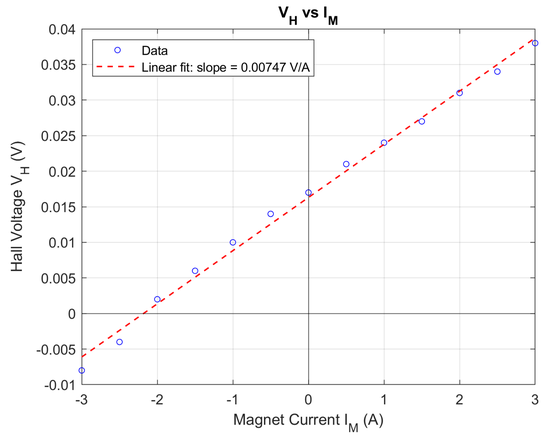

Figure 4: Hall voltage V_H vs. magnet current I_M.

Hall Voltage vs. Magnetic Field The linear fit determined carrier density. The offset from the origin is consistent with Seebeck-related thermoelectric shifts.

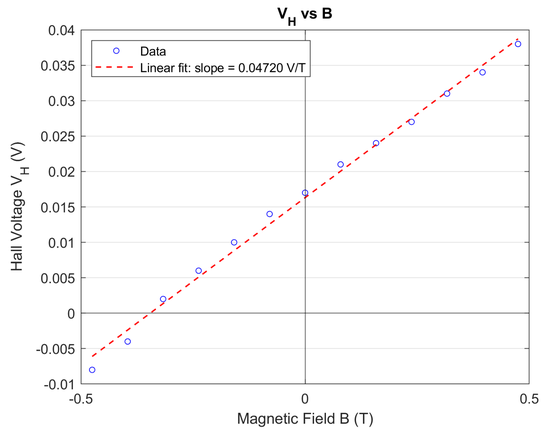

Figure 5: Hall voltage V_H vs. magnetic field B.

Final Transport Properties

Contact Pair	ρ (Ω·cm)	μ (cm^2/Vs)	Accepted μ_h
2–3	7.82 ± 0.21	313.74 ± 12.01	450–500
5–6	7.54 ± 0.20	325.62 ± 12.47	450–500

Carrier Density $(2.544 \pm 0.071) \times 10^{15}\ \mathrm{cm}^{-3}$